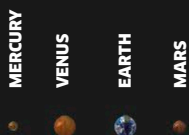


Size comparison

With a diameter of nearly 870,000 miles (1.4 million km), the sun is 10 times wider than Jupiter, the biggest of the planets, and over 1,000 times more massive.

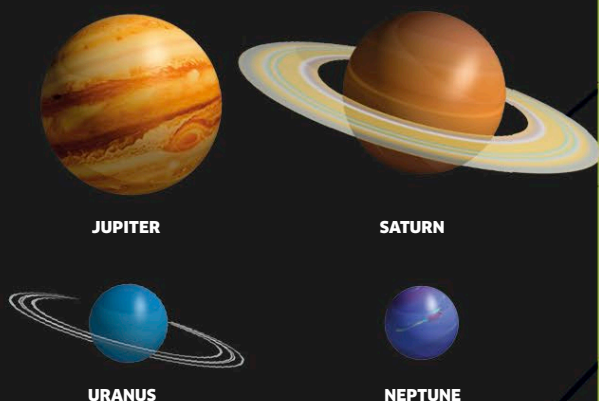
Inner planets

The inner four planets are smaller than the outer four. They are called terrestrial planets.



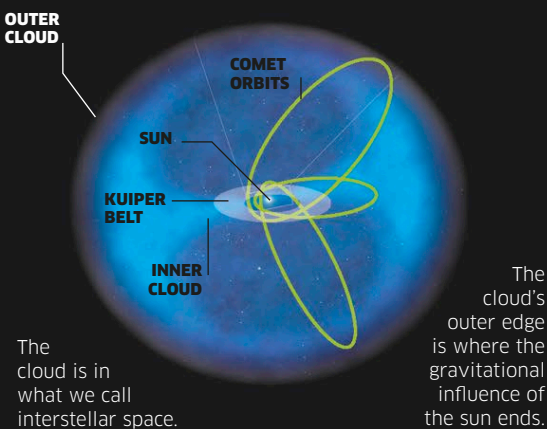
Outer planets

The outermost four planets are larger and made up of gas, so they are called the gas giants.



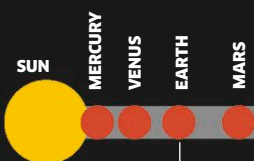
Oort Cloud

The Oort Cloud is a ring of tiny, icy bodies that is thought to extend between 50,000 and 100,000 times farther from the sun than the distance from the sun to Earth—but it's so far away that no one really knows.



Distance from the sun

It is hard to imagine how far Earth is from the sun, and how much bigger the sun is than Earth. If Earth were a peppercorn, the sun would be the size of a bowling ball—100 times bigger.



Earth is 92.9 million miles (149.6 million km) from the sun—or one astronomical unit (AU).

JUPITER

Jupiter is 484 million miles (780 million km) from the sun, which is equal to 5.2 AU.

SATURN

Saturn orbits on average 890 million miles (1.43 billion km) from the sun, or 9.58 AU.

Kuiper Belt

The Solar System does not end beyond Neptune: the Kuiper Belt (30–55 AU from the sun) is home to smaller bodies that include dwarf planets.

Neptune

Astronomers predicted the existence of the blue planet by its effect on the orbit of Uranus.

Uranus

The icy blue giant rotates on its side as it orbits the sun. Winter on the planet lasts 42 years.

Saturn

The second largest planet, Saturn has 62 moons and is circled by sparkling fragments of ice that form its rings.

Jupiter

More massive than the other planets combined, Jupiter rotates once every 10 hours, whipping its red clouds into stripes and swirling storms.

Comets

These icy bodies develop spectacular tails of gas and dust as they near the sun.

Orbits

The orbits of the planets and most asteroids around the sun are aligned. Comets, though, can orbit at any angle.

Orbiting planets

There are eight planets in the solar system. They form two distinct groups. The inner planets—Mercury, Venus, Earth, and Mars—are solid balls of rock and metal. The outer planets—Jupiter, Saturn, Uranus, and Neptune—are gas giants: enormous, swirling globes made mostly of hydrogen and helium.

The Solar System

The solar system is a huge disk of material, with the sun at its center, that stretches out over 19 billion miles (30 billion km) to where interstellar space begins.

Most of the solar system is empty space, but scattered throughout are countless solid objects bound to the sun by gravity and orbiting around it. These include the eight planets, hundreds of moons and dwarf planets, millions of asteroids, and possibly billions of comets. The sun itself makes up 99.8 percent of the mass of the solar system.